**Q1 Report: Speech Signal Processing and Analysis**

methods including:

- MFCC (Mel-Frequency Cepstral Coefficients)
- Spectral leakage and Signal-to-Noise Ratio (SNR)
- Voiced vs unvoiced speech detection
- Phonetic mapping using a pre-trained model

**Methodology**

**MFCC Extraction**

MFCCs are widely used features in speech processing as they approximate the human auditory system.

**Steps:**

1. Load audio signal
2. Frame the signal
3. Apply Hamming window
4. Compute FFT and power spectrum
5. Apply Mel filter banks
6. Take log energies
7. Apply Discrete Cosine Transform (DCT)

**Output:**

- MFCC matrix of shape **(frames × coefficients)**
- Example: (199 × 13)

**2.3 Voiced vs Unvoiced Detection**

Speech signals are classified into:

- **Voiced**: periodic (e.g., vowels)
- **Unvoiced**: noise-like (e.g., consonants)

**Method:**

- Frame-wise energy calculation
- Threshold-based classification

**Results:**

- Voiced Frames: 40
- Unvoiced Frames: 159

**Phonetic Mapping**

A pre-trained **Wav2Vec2 model** is used for phonetic transcription.

**Steps:**

1. Load audio
2. Pass through Wav2Vec2 model
3. Decode output tokens

**Observation:**

- Input signal was a **synthetic sine wave**
- Model produced **no meaningful transcription**

**RMSE Calculation:**

Used to compare predicted vs reference signal.

RMSE = 0.7177

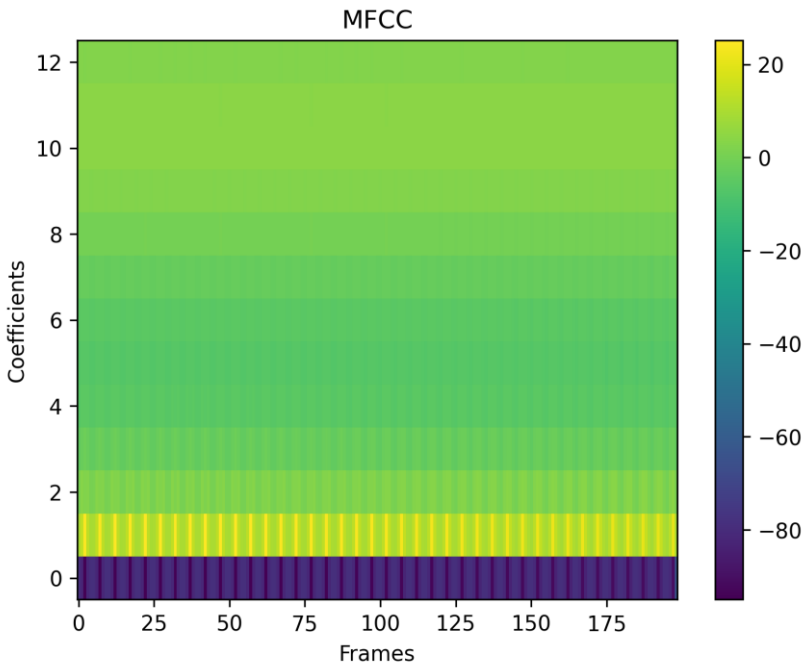
**Interpretation:**

High RMSE indicates poor phonetic alignment due to non-speech input.

**Results and Visualizations**

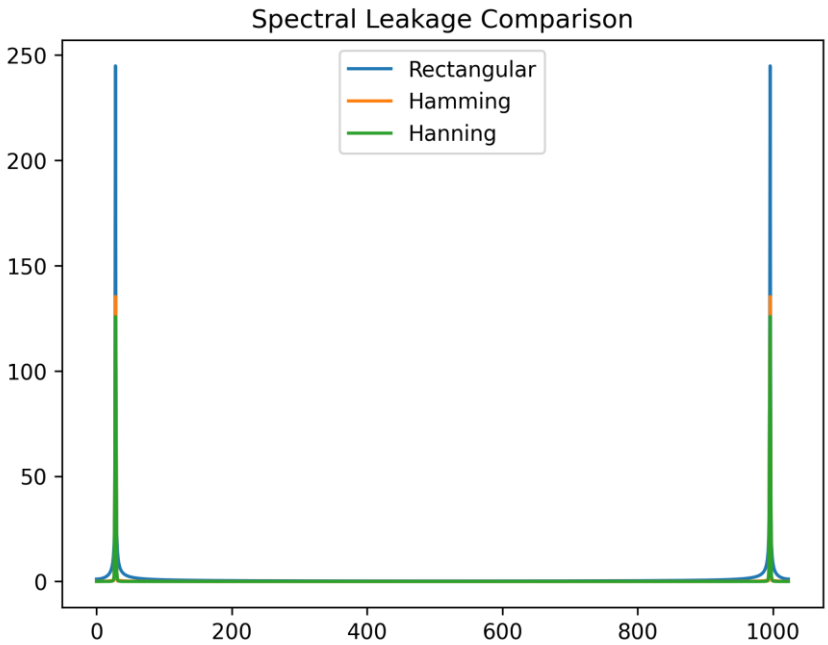
**MFCC Plot**

- Displays time vs frequency coefficients
- Shows energy concentration in lower frequencies



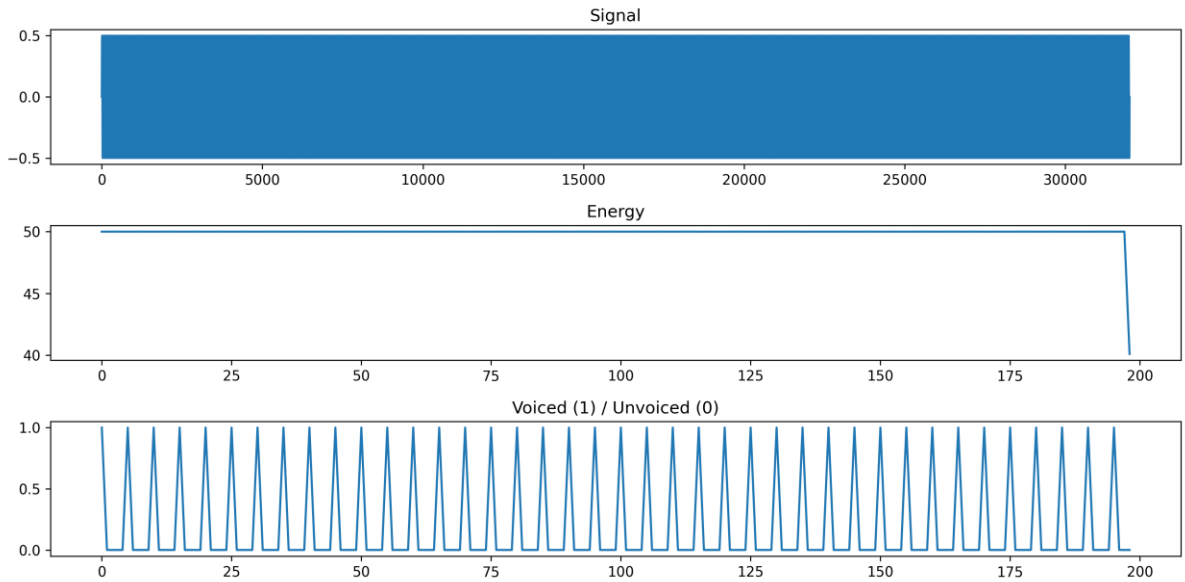
**Spectral Leakage Comparison**

- Visual comparison of window functions
- Hamming and Hanning perform better




**3.3 Voiced/Unvoiced Plot**

- Frame classification visualization




**RMSE Summary**

| Metric | Value |
|---|---|
| RMSE | 0.7177 |

- MFCC effectively captures speech characteristics
- Window functions reduce spectral leakage
- Simple energy-based methods can classify voiced/unvoiced frames
- Deep models like Wav2Vec2 require real speech input for meaningful results

This section demonstrated key speech processing techniques and their practical implementation. While MFCC and spectral analysis performed as expected, phonetic mapping highlighted the importance of using realistic speech data. These foundational techniques are essential for building advanced speech systems.